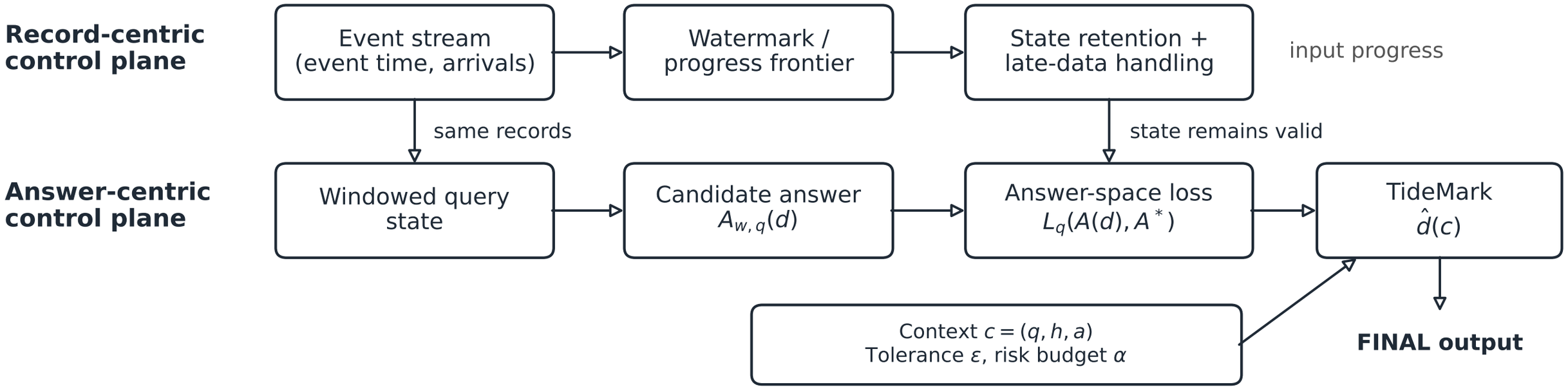
Record-centric control plane
Event stream (event time, arrivals)
Watermark / progress frontier
State retention + late-data handling
input progress
same records
state remains valid
Answer-centric control plane
Windowed query state
Candidate answer $A_{w,q}(d)$
Answer-space loss $L_q(A(d), A^*)$
TideMark $\hat{d}(c)$
Context $c = (q, h, a)$
Tolerance ε, risk budget α
FINAL output